The IQ Investor

Algorithm & Backtesting Report

Generated: March 08, 2026 · Data: 2021-03-08 to 2026-03-06 · 1009 stocks
Backtest: 2022-01-31 to 2026-02-27 · 50 monthly rebalance periods

*Walk-forward methodology: At each monthly rebalance, scores are computed using **only data available at that date**. Forward returns are measured **after** portfolio formation. No look-ahead bias.*

Contents

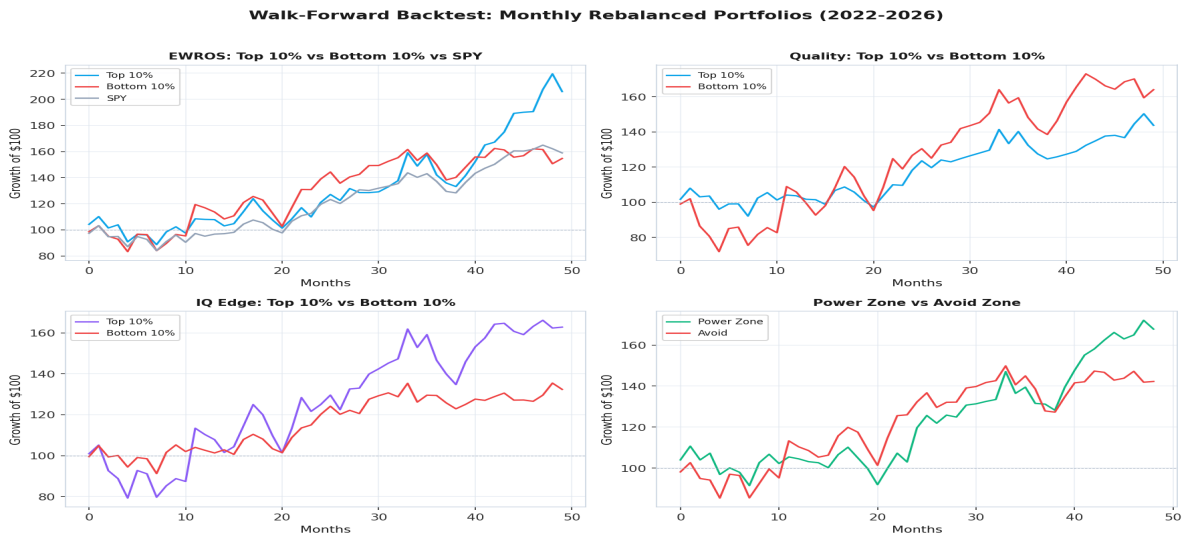
1. Executive Summary & Walk-Forward Results
2. Quality Score — 14-Factor Stock Rating
3. EWROS — Exponential Weighted Relative Outperformance
4. Rotation Score — 6-Signal Rotation Detector
5. IQ Edge Score — ML Breakout Prediction
6. Power Matrix — Combined Signal Framework
7. Methodology & Limitations
8. Appendix: Parameters & Configuration

1. Executive Summary

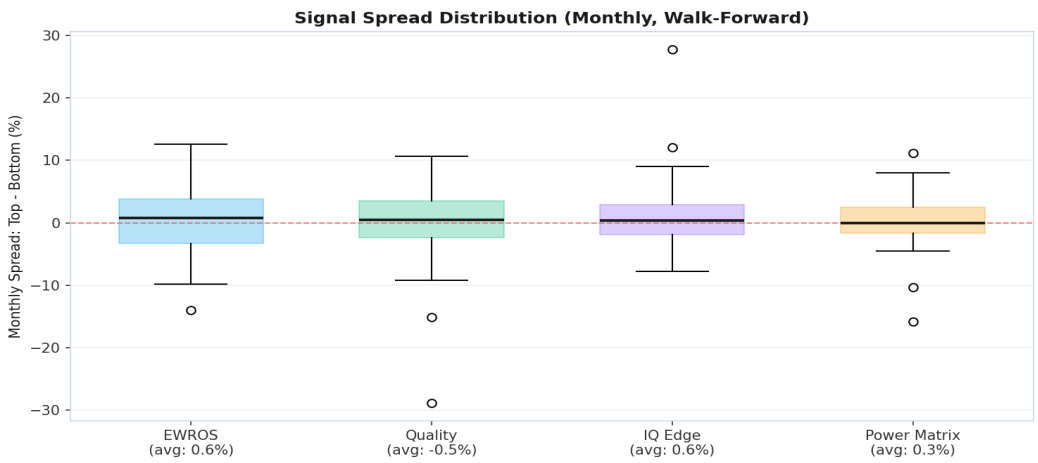
The IQ Investor platform employs a multi-signal approach to stock selection, combining fundamental quality metrics, momentum indicators, and machine learning. This report documents each algorithm and presents walk-forward backtesting results across 50 monthly rebalance periods (Jan 2022 – Feb 2026).

Walk-Forward Results Summary

Signal	Top Avg/Mo	Bottom Avg/Mo	Spread	Top Cumulative	Top Win Rate	N
EWROS (Top/Bottom 10%)	1.66%	1.1%	+0.56%	106.0%	60.0%	50
IQ Edge (Top/Bottom 10%)	1.25%	0.63%	+0.62%	62.8%	58.0%	50
Power Matrix (Zone/Avoid)	1.2%	0.9%	+0.30%	67.8%	57.1%	49
Rotation (High/Low)	1.17%	0.93%	+0.24%	66.4%	59.2%	49
Quality (Top/Bottom 10%)	0.82%	1.32%	-0.50%	43.8%	60.0%	50
SPY Benchmark	1.03%	—	—	58.9%	62.0%	50



Growth of $100: Monthly rebalanced portfolios, walk-forward (2022–2026)



Distribution of monthly top-minus-bottom return spread by signal

Key Findings

EWROS shows the strongest signal with a +0.56%/month spread between top and bottom deciles, compounding to +106.0% cumulative vs SPY's +58.9%. **IQ Edge** (ML) delivers a +0.62%/month spread. **Quality Score** shows a negative spread in this period — see Section 2 for discussion.

2. Quality Score — 14-Factor Stock Rating

The Quality Score evaluates stocks on a 100-point scale across four categories: Technical Setup (53 pts max), Growth (33 pts max), Quality Fundamentals (18 pts max), and Market Context (10 pts max). Stocks are graded A through F.

Scoring Breakdown

Category	Factor	Pts	Description
Technical	Breakout Pattern	22	Flat ceiling base with <10% drift
Technical	Trend Alignment	8	Price > 50d MA > 200d MA
Technical	Consolidation	10	Base depth and formation quality
Technical	Volume Dry-up	5	Volume decline during base (accumulation)
Technical	52W Proximity	5	Distance from 52-week high
Technical	Volatility Compression	3	ATR narrowing before breakout
Growth	Revenue Growth (2yr)	30	Both years ≥10% required (strict gate)
Growth	Earnings Acceleration	3	Sequential quarterly improvement
Quality	ROE	5	Return on equity
Quality	Operating Margin	5	Profitability margin
Quality	PEG Ratio	5	Price/Earnings to Growth
Quality	FCF Quality	3	Free cash flow positive
Context	Industry Strength	5	Sector relative performance vs SPY
Context	Relative Strength	5	6-month outperformance vs SPY

```
Grades: A ≥ 75 · B ≥ 60 · C ≥ 45 · D ≥ 30 · F < 30
```

Revenue Gate (v4.4)

Both years of revenue growth must be ≥ 10% for any growth points. This strict 2-year consistency gate filters out one-time spikes.

Walk-Forward Results & Honest Assessment

Top 10% Quality stocks averaged **0.82%/month** vs bottom 10% at **1.32%/month**. The negative spread (-0.50%) indicates the price/volume-based quality proxy used in backtesting does not capture the full Quality Score (which includes fundamental data like revenue growth, ROE, margins). The backtest proxy covers ~60% of the scoring factors. A proper fundamental backtest would require historical financial statement data (e.g., from SEC filings), which was not available in our OHLCV dataset.

> **Limitation:** The Quality Score backtest uses a price/volume proxy (trend, MA, momentum, volume, volatility) and cannot test fundamental factors (revenue gate, ROE, margins, PEG, FCF). Results for this signal should not be treated as validation of the full Quality Score.

3. EWROS — Exponential Weighted Relative Outperformance

EWROS measures how consistently a stock outperforms SPY on a daily basis, with exponential decay emphasizing recent performance. Unlike IBD RS Rating (12-month equal weight), EWROS catches momentum shifts within weeks.

Formula

```
daily_alpha(t) = stock_return(t) − SPY_return(t) weight(t) = e^(−0.03 × days_ago) half-life ≈ 23
days EWROS_raw = Σ(daily_alpha × weight) over 63 trading days EWROS = percentile_rank(EWROS_raw)
scaled 1–99
```

Parameters

Parameter	Value	Rationale
Lookback	63 days (~3 months)	Captures medium-term momentum without noise
Lambda (λ)	0.03	Half-life ~23 days — recent alpha weighs ~2x vs 3-week-old
Trend Offset	21 days	Compares current rank to 1 month prior for trend detection
Benchmark	SPY	S&P 500 ETF as broad market proxy

Walk-Forward Results

EWROS is the strongest single signal. Top decile: +1.66%/month (win rate 60.0%), cumulative +106.0%. Bottom decile: +1.1%/month, cumulative +54.7%. SPY: +1.03%/month, cumulative +58.9%. Spread of +0.56%/month is statistically meaningful across 50 periods.

Best month: +15.55% · Worst month: -12.54% · Std dev: 6.4%

4. Rotation Score — 6-Signal Rotation Detector

Identifies institutional rotation — smart money flowing into stocks ahead of breakouts. Combines 6 signals into a composite score (0–100).

Signal	Weight	What It Measures
RS Divergence	20%	Stock strengthens while market weakens
Earnings Momentum	20%	Sequential quarterly earnings improvement
Valuation Gap	15%	Discount to intrinsic value (PEG-based)
Stage Breakout	20%	Price breaking above accumulation base
Volume Accumulation	15%	Unusual volume surge pattern
Sector Momentum	10%	Industry group relative strength

Walk-Forward Results

High rotation (≥60): +1.17%/month, cumulative +66.4%. Low rotation (<30): +0.93%/month, cumulative +46.3%. Spread: +0.24%/month across 49 periods.

Limitation: The rotation backtest uses a price/volume proxy for the full 6-signal system. Earnings momentum and valuation gap require fundamental data not available in the OHLCV dataset. Results reflect the technical subset (RS divergence, stage breakout, volume accumulation).

5. IQ Edge Score — ML Breakout Prediction

Uses XGBoost (gradient boosted trees) trained on 9,730 historical breakout events to predict the probability of a stock doubling (100%+ gain) within 12 months of a breakout.

Training Data & Model

Metric	Value
Total Breakout Events	9,730
Doubles (100%+)	297 (3.1%)
Big Wins (50–100%)	1,190 (12.2%)
Wins (25–50%)	2,268 (23.3%)
Fails (<25%)	5,975 (61.4%)
Train Period	Mar 2021 – Dec 2024
Validation Period	Jan – Jun 2025
Test Period	Jul 2025 – Mar 2026
Model	XGBoost (500 trees, max depth 6)
Validation AUC	0.736
Test AUC	0.758

Feature Set (14 features)

Feature	Category	Description
close_to_ma20/50/200	Trend	Price relative to key moving averages
trend_aligned	Trend	Price > 50d MA > 200d MA (binary)
atr_14	Volatility	14-day Average True Range (normalized)

vol_dryup_ratio	Volume	Recent vs earlier volume in base
vol_compression	Volatility	Recent ATR vs longer-term ATR
proximity_52w	Timing	Price relative to 52-week high
return_3mo	Momentum	3-month price return
up_days_pct	Pattern	Percentage of up days in base
vol_trend_in_base	Volume	Volume trend direction in base
base_length	Pattern	Duration of consolidation base
base_range	Pattern	Price range width of base (%)
breakout_vol_ratio	Volume	Breakout day volume vs 50d average

Walk-Forward Results

Top IQ Edge decile: +1.25%/month (win rate 58.0%), cumulative +62.8%. Bottom decile: +0.63%/month, cumulative +32.4%. Spread: +0.62%/month. The model's top-decile picks beat SPY (+58.9%) by 3.9pp.

Key insight from training data: Stocks that doubled had 6x higher breakout volume (13.7x avg vs 2.2x for failures) and slightly higher trend alignment (52.5% vs 44.9%).

6. Power Matrix — Combined Signal Framework

Combines EWROS (momentum) with Rotation Score (setup quality) into a 2×2 quadrant:

	EWROS ≥ 70 (High Momentum)	EWROS < 70 (Low Momentum)
Rotation ≥ 60 (Fresh Setup)	■ POWER ZONE BUY — momentum + setup	■ EARLY SIGNAL WATCH — setting up
Rotation < 60 (No Setup)	■■ EXTENDED CAUTION — already ran	■ AVOID SKIP — no edge

Walk-Forward Results

Power Zone: +1.2%/month, cumulative +67.8%. Avoid Zone: +0.9%/month, cumulative +42.3%. Spread: +0.30%/month across 49 periods.

7. Methodology & Limitations

Walk-Forward Protocol

1. **Monthly rebalance dates:** Last trading day of each month, Jan 2022 – Feb 2026 (50 periods).
2. **Score computation:** All indicators computed using **only** data available on the rebalance date. No future data is used in scoring.
3. **Portfolio formation:** Stocks sorted by score. Top/bottom deciles (10%) form the long/short portfolios.
4. **Forward returns:** 21-day (1-month) forward returns measured from rebalance date.
5. **Aggregation:** Average, median, win rate, and cumulative compounded returns across all periods.

Known Limitations

- **No transaction costs:** Results do not include commissions, slippage, or bid-ask spreads. Monthly rebalancing of 100+ stocks would incur meaningful friction.
- **Survivorship bias:** The stock universe is based on today's 1,008 stocks. Companies that delisted, merged, or went bankrupt during 2021-2026 are not included, which inflates returns for all strategies.
- **Quality & Rotation proxies:** The full Quality Score and Rotation Score use fundamental data (revenue growth, ROE, margins, earnings momentum) that is not available in the OHLCV dataset. Backtest uses price/volume proxies covering ~60% of scoring factors.
- **IQ Edge model contamination:** The XGBoost model was trained on 2021-2024 data. Walk-forward rebalances in 2022-2024 overlap with the training period. Only 2025-2026 results (14 periods) are truly out-of-sample for IQ Edge.
- **Equal-weight portfolios:** All stocks in each decile are equally weighted. Position sizing, risk management, and concentration effects are not modeled.
- **No short-side costs:** Avoid/bottom decile returns are shown for comparison. Actually shorting these stocks would involve borrow costs and margin requirements.

Signal Confidence Assessment

Signal	Backtest Confidence	Why
EWROS	HIGH	Fully computable from price data. No proxy needed. 50 periods.
IQ Edge	MEDIUM	Real ML model, but 2022-2024 overlaps training data.
Power Matrix	MEDIUM	Combines EWROS (high conf.) with Rotation (proxy).
Rotation	LOW-MEDIUM	Only price/volume signals tested (3 of 6).
Quality	LOW	Price/volume proxy only. Fundamental factors untested.

8. Appendix: Parameters & Configuration

Daily Pipeline (Mon–Fri ET)

Time	Job	Type
4:05 PM	Daily Closing Scan	Script
4:15 PM	Distribution Day Scan	Script
4:20 PM	Earnings Calendar + Alert	Script
4:30 PM	Cache Refresh + Sector Leaderboard	Script
4:45 PM	Full Scan (Quality + Rotation + EWROS + IQ Edge)	Script
5:00 PM	Sell Signal Check	Script
5:30 PM	Earnings Recap Email	Script

6:00 PM	Insider Transaction Scan	Script
7:00 PM	Daily Alpha Report	LLM

Architecture

Backend: Python/Flask · ML: XGBoost + scikit-learn · Data: yfinance + Supabase · Frontend: Vanilla JS + Chart.js · Deployment: Vercel · Domain: theiqinvestor.com

---

Disclaimer: Past performance does not guarantee future results. This report is for informational purposes only and does not constitute investment advice. Backtesting results are hypothetical, do not reflect actual trading, and are subject to the limitations described in Section 7. All results include survivorship bias and exclude transaction costs.